The Transmission Electron Microscope (TEM) was the first type of Electron Microscope to be developed.

2.3.1 Principle

The fundamental principle of electron microscope is similar to light microscope. In electron microscope, a high velocity beam of electrons is used instead of photons. In the electron gun, electrons are emitted from the surface of the cathode and accelerated towards the anode by high voltage to form a high energy electron beam. All lenses in the electron microscope are electromagnetic. Charged electrons interact with the magnetic fields and magnetic force focuses an electron beam. The condenser lens system controls the beam diameter and convergence angles of the beam incident on a specimen. The image is formed either by using the transmitted beam or by using the diffracted beam. The image is magnified and focused onto an imaging device, such as a fluorescent screen, on a layer of photographic film, or to be detected by a sensor.

Sample Preparation

Preparation of specimens is the most complicated and skillful step in EM. The material to be studied under electron microscopy must be well preserved, fixed, completely dehydrated, ultrathin and impregnated with heavy metals that sharpen the difference among various organelles.

The material is preserved by fixation with glutaraldehyde and then with osmium tetroxide. The fixed material is dehydrated and then embedded in plastic (epoxy resin) and sectioned with the help of diamond or glass razor of ultra-microtome.

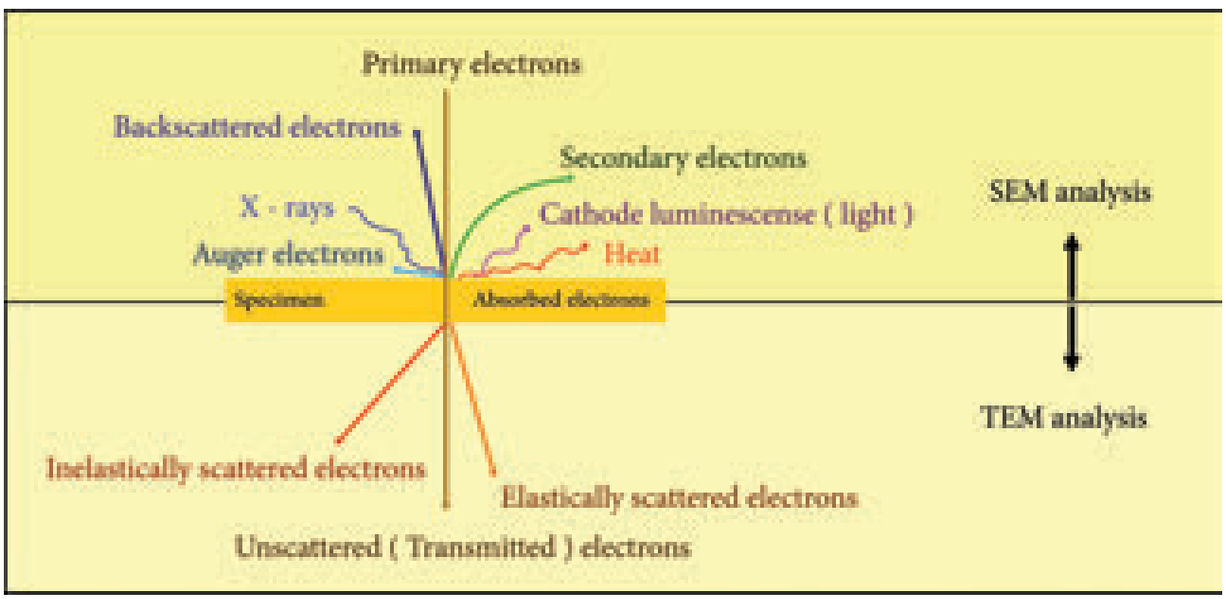


Figure 2.8: Interaction between electron beams with specimen

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